Terrain detection and segmentation for autonomous vehicle navigation

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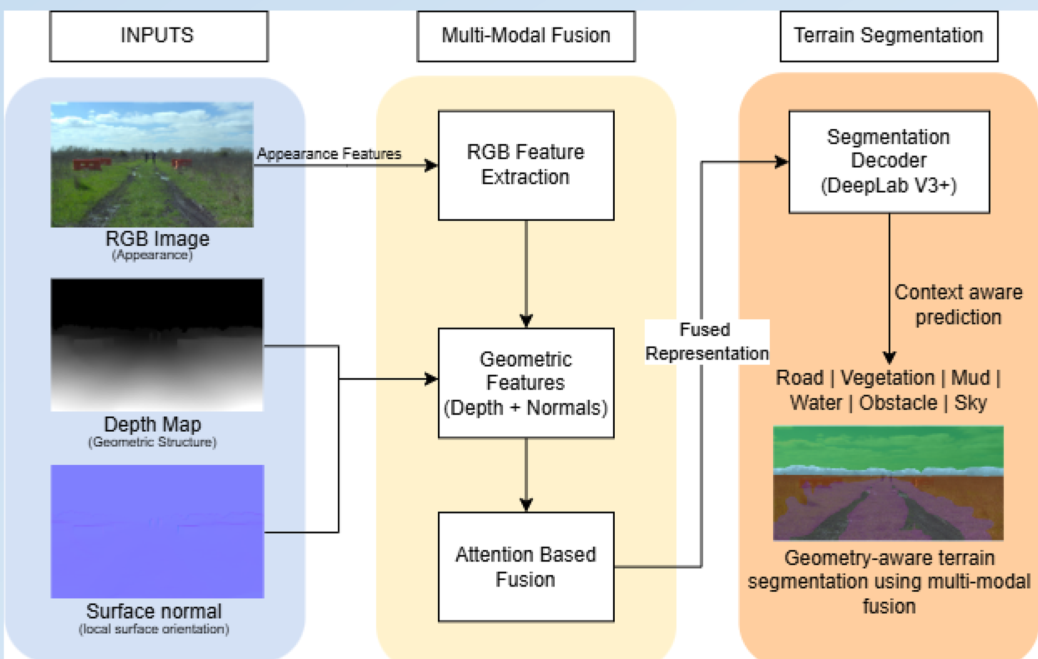
Graphical Abstract



•Multi-modal fusion improves terrain understanding in off-road scenes
•Depth and normals provide geometric cues beyond RGB
•Enables robust terrain segmentation for autonomous navigation

Background

• Challenge in Off-road Terrain Perception:
Autonomous vehicles operating in unstructured environments face large appearance variations due to lighting, weather, and terrain diversity. RGB-only segmentation models often fail to reliably distinguish visually similar surfaces such as mud, water, and gravel.

• Limitations of RGB-based Segmentation
Purely appearance-driven models are highly sensitive to domain shift and illumination changes across datasets. This limits their robustness and generalization when deployed beyond controlled urban scenes.

• Motivation for Multi-modal Learning
Geometric cues such as depth and surface normals provide stable structural information independent of appearance. Fusing RGB with depth and normals enables improved terrain boundary detection.

(a) Structured Environments (b) Unstructured Environments

Literature Review

Ref	Year	Method / Dataset	Key insights
[1]	2024	Systematic review	Highlights domain shift
[2]	2021	Off-road segmentation	Works in unstructured terrain
[3]	2024	Depth-assisted segmentation	Geometry improves robustness
[4]	2016	Cityscapes Dataset	Urban benchmark
[5]	2020	Rellis-3D	Rural benchmark

Objective and Proposed Methodology

- A **multi-modal terrain segmentation framework** is presented for autonomous navigation in unstructured environments, targeting robustness under lighting variation and terrain ambiguity. The approach integrates **RGB appearance**, **depth geometry**, and **surface normals (Trifecta)** to capture complementary visual and geometric cues.

- A **DeepLabV3 architecture with a MobileNetV3 backbone** is employed for efficient pixel-wise terrain segmentation.

- Three modality configurations are evaluated:
  - RGB: Visual appearance information.
  - RGB-D: RGB combined with depth maps to capture scene geometry.
  - Trifecta (RGB + Depth + Normals): Depth-derived surface normals are added to encode local surface orientation.

Cross-dataset evaluation on **Cityscapes, KITTI, and RELLIS-3D** is used to assess generalization under domain shift.

Methodology

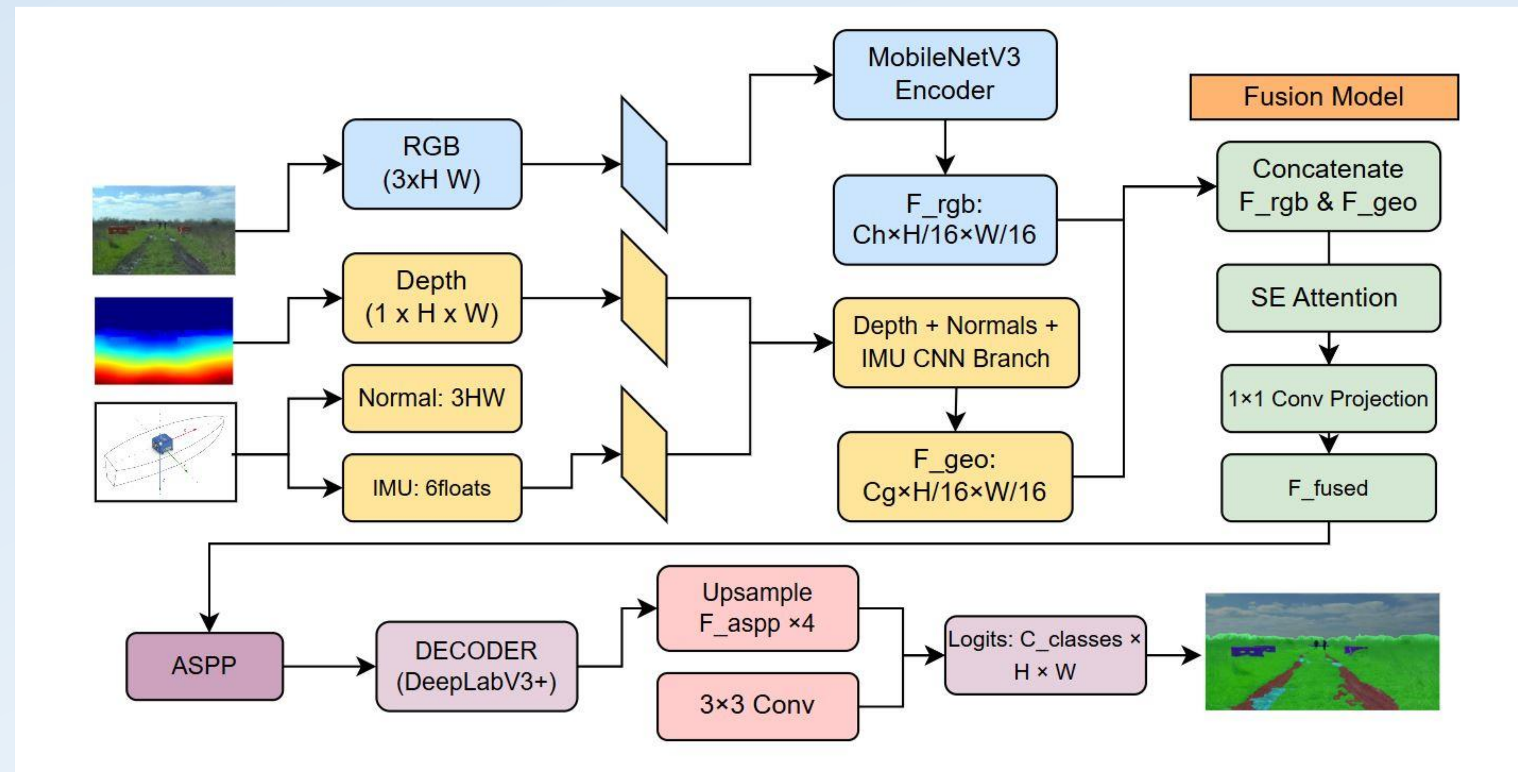


Fig : Multi-modal fusion architecture combining RGB appearance and geometric cues using a DeepLabV3+ backbone for terrain segmentation.

Results and Discussion

In-Dataset Performance
The segmentation performance was evaluated using mean Intersection-over-Union (mIoU) across three datasets.

Variant	Modalities	Cityscapes mIoU	KITTI mIoU	RELLIS-3D mIoU
A	RGB	0.8696	0.934	0.7947
B	RGB + Depth	0.8697	0.922	0.752
C	RGB + Depth + Normals	0.8671	0.928	0.755

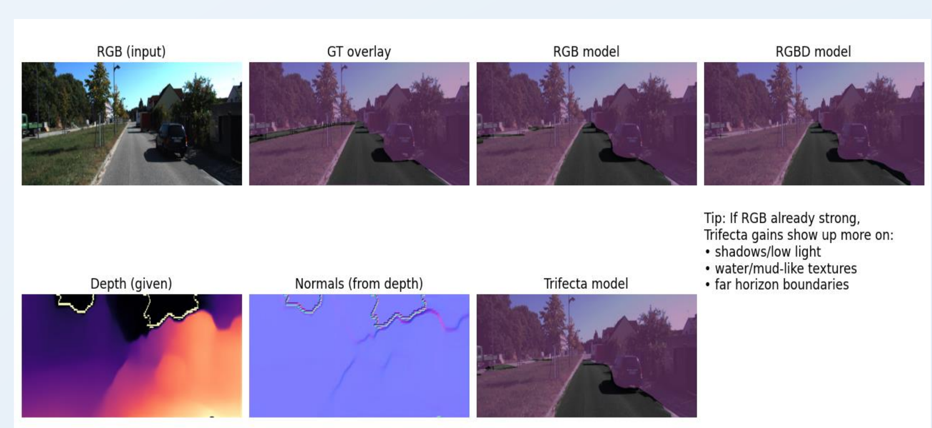
The Trifecta model consistently demonstrates improved or competitive performance, particularly in unstructured datasets such as RELLIS-3D, where geometric cues play a critical role.

Cross-Dataset Generalization
Evaluation highlights the robustness of each modality under domain shift.

Train → Test	RGB	RGB-D	Trifecta
KITTI → RELLIS-3D	Strong	Moderate	Very Strong
RELLIS-3D → KITTI	Moderate	Weak	Moderate

The Trifecta configuration exhibits superior generalization, confirming that surface normals help preserve structural consistency across domains.

Qualitative Results
It demonstrate that the Trifecta model produces cleaner boundaries, fewer false positives, and better segmentation of visually ambiguous terrain regions.



Conclusion and Future Work

- Multi-modal fusion significantly improves terrain segmentation robustness.
- Surface normals complement depth and RGB by capturing local geometric orientation.
- Trifecta demonstrates superior cross-dataset generalization, making it suitable for real-world autonomous navigation.

Future Work
- Extend to multi-class terrain segmentation
- Incorporate temporal consistency for video streams
- Deploy on real-time embedded platforms

References

[1] M. M. Kabir, J. R. Jim, and Z. Istenes, “Terrain detection and segmentation for autonomous vehicle navigation: A state-of-the-art systematic review,” Information Fusion, Aug. 2024.
[2] A. Shaban, X. Meng, J. Lee, B. Boots, and D. Fox, “Semantic terrain classification for off-road autonomous driving,” in Conf. Robot Learning (CoRL), London, U.K., 2021.
[3] S. Li, A. Gupta, and K. Ramachandran, “Semantic segmentation with depth vision for off-road terrain understanding,” IEEE Trans. Intell. Transp. Syst.,vol. 25, no. 6, 2024.
[4] M. Cordts et al., “The Cityscapes dataset for semantic urban scene understanding,” in Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR), 2016, pp. 3213–3223.
[5] P. Jiang, P. Osteen, M. Wigness, and S. Saripalli, “RELLIS-3D Dataset: Data, Benchmarks and Analysis,” arXiv:2011.12954, 2020.